

Sound Localising Methods

1. Model the incoming sound

First we have to model the incoming sound somehow.

$$x_i(t) = s(t) * h_i(t) + n_i(t) \quad (1)$$

Where:

- $x_i(t)$ is the signal captured by the i -th microphone.
- $s(t)$ is the source signal.
- $h_i(t)$ is the room response. It consists of the time difference that the sound has to travel and all other reverberations. (Convolution can apply these features at the same time).
- $n_i(t)$ is the noise captured by the i -th microphone.

2. Cross-correlation

Then basically we'll calculate cross-correlation (later we'll use tricks but the core is still a cross-correlation problem).

So:

$$R_{ij}(\tau) = \int_0^T x_i(t)x_j(t + \tau) dt \quad (2)$$

In theory we could do this but when we're working with computers we should apply discrete methods.

$$R_{ij}[\tau] = \sum_{k=0}^N x_i[k]x_j[k + \tau] \quad (3)$$

It is a cross correlation that gives us back how the two signals match as a function of τ , then we can select the τ that maximizes this function and say that's the time difference of arrival.

3. Frequency Domain

But we can do it in a way more sophisticated way by looking at these events in frequency domain.

But before we do that it's worth noticing that the computation of cross-correlation and convolution is really similar. And convolution in time domain becomes multiplication in frequency domain.

So:

$$R_{ij} = x_i(t) \star x_j(-t) \quad (4)$$

(The minus sign becomes complex conjugation).

$$G_{ij}(\omega) = X_i(\omega)X_j^*(\omega) \quad (5)$$

Before we transform back to get the TDOA we whiten the signals by removing the magnitude. This makes the method more precise.

So we'll transform $\frac{G_{ij}}{|G_{ij}|}$ back to time domain.

$$R_{ij}^{PHAT}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{X_i(\omega)X_j^*(\omega)}{|X_i(\omega)X_j^*(\omega)|} e^{i\omega\tau} d\omega \quad (6)$$

So by doing this we'll get a pulse at the estimated TDOA.

So with GCC-PHAT we can get the time delays for every pair of mics.